

Muhammad Usaid

SOFTWARE ENGINEER

+92-334-4619036

usaidehsan01@gmail.com

[LinkedIn Profile](#)

[GitHub Profile](#)

Lahore, Pakistan

CAREER SUMMARY

FAST University graduate with a focus on Data Science and Web Development. Passionate about building practical solutions and continuously refining my skills. Enjoy tackling challenges, exploring new technologies, and staying up to date with industry trends. Always eager to collaborate, learn, and contribute to meaningful projects.

EDUCATION

National University of Computer and Emerging Sciences

Sep 2020 - June 2024

Bachelor of Computer Science

Lahore

PROFESSIONAL EXPERIENCE

Software Engineer / Txend Inc

September 2024 - Present

- Designed, developed, and managed end-to-end data pipelines to ingest and process data from Dynamics CRM, social media, REST APIs, BigQuery and file-based sources.
- Led the implementation of AWS-based ETL workflows, utilizing services like EventBridge, Lambda, S3, Redshift, EMR, SageMaker, and Transfer Family for efficient data processing, storage, and scheduling.
- Analyzed collected data from scheduled pipelines to generate KPIs, uncover trends, and provide actionable insights for data-driven decision-making.

AI/ML Engineer / Paklogics

March 2024 - August 2024

- Developed efficient bots and web scrapers using Selenium for automated data extraction and task automation, optimizing workflow efficiency.
- Integrated ChatGPT and ChatGroq with LangChain, implementing prompt engineering and database storage to maintain user-specific chat histories for improved contextual responses.
- Enhanced automation reliability by leveraging Recaptcha_Solver, pynput, DaisySMS, and Oxylabs proxies to bypass restrictions and ensure seamless execution.

Associate Software Engineer / Devoin Labs

November 2023 - February 2024

- Optimized AI chatbot responses for accurate Python and multi-language coding inquiries by refining prompt engineering techniques.
- Assisted in training AI models with enhanced data visualization using Matplotlib, Seaborn, and Plotly to improve interpretability and insights.
- Gaining hands-on experience in the MERN stack (MongoDB, Express.js, React, Node.js) through project development while contributing to the design and maintenance of the company's website.

Machine Learning Intern / Ebryx

June 2023 - Sep 2023

- Developed and deployed machine learning and deep learning models using Scikit-Learn, PyTorch, TensorFlow, and Keras to analyze user behavior and identify potential insider threats.
- Engineered feature extraction and preprocessing pipelines to handle large-scale security data, optimizing model performance for anomaly detection.
- Designed and implemented Python-based security monitoring scripts to collect, process, and analyze Linux device activity, leveraging cybersecurity principles to detect unusual patterns and enhance threat detection.

PROJECTS

Enterprise Data Warehouse & Analytics on AWS

- Designed and implemented a data warehouse to centralize data from multiple sources, including Dynamics CRM, social media, REST APIs, and file-based systems.
- Developed AWS-based ETL pipelines utilizing EventBridge for scheduling, Lambda for processing, S3 for storage, Redshift for warehousing, and EMR/SageMaker for data transformation, spark jobs and analytics.
- Performed data analysis on the aggregated data, leveraging AWS services to compute KPIs and derive actionable business insights.

Context-Aware AI Chatbot with RAG & Embedding-Based Retrieval

- Developed a Retrieval-Augmented Generation (RAG) based AI chatbot that learns user behavior, preferences, and habits to provide personalized and context-aware responses.
- Implemented embedding-based retrieval using Pinecone for efficient similarity search and memory retention, enabling dynamic and adaptive learning.
- Built a FastAPI-powered backend leveraging Transformers and Ollama for response generation, with a React Native frontend for real-time, seamless user interaction across devices.

Remote Online Notary System

- Developing a Remote Online Notary (RON) app prototype using the SERN stack (SQL, Express.js, React, Node.js) with audio-visual session recording for audit compliance.
- Implementing AI-driven identity verification using Computer Vision and securing data with tamper-evident storage via ImmuDB.
- Ensuring document authenticity and integrity through digital signatures, leveraging asymmetric cryptography for secure notarization.

Tinder and Snapchat Automation Bot

- Built an automation bot using Selenium to handle account creation, login, and OTP verification for Tinder and Snapchat, streamlining onboarding and access.
- Developed independent Tinder swiping automation while integrating OpenAI's chatbot for automated message replies and conversation flow management on both platforms.
- Implemented MongoDB for storing chat histories, ensuring structured data retention for future analysis and user interaction tracking.

Rock-Paper-Scissors Hand Gesture Detection System

- Developed an AI-powered Rock-Paper-Scissors game in Python using Keras and TensorFlow for real-time hand gesture recognition.
- Trained a Convolutional Neural Network (CNN) on a Kaggle-sourced image dataset to accurately classify hand gestures.
- Integrated the model with a live camera feed to enable real-time gameplay and dynamic user interaction.

Anomaly Detection in Electrocardiogram (ECG) Signals

- Implemented LSTM autoencoders to analyze ECG signals, accurately reconstructing normal heartbeats for anomaly detection.
- Leveraged reconstruction error thresholds to identify irregular heart rhythms and potential cardiac abnormalities.

TECHNICAL SKILLS

- **Programming Languages:** C/ C++, Python, Ruby, R
- **Web Technologies:** HTML, CSS, JavaScript, React, Node JS, Express JS, Firebase, FastAPI, React Native
- **Data Science Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, LangChain, Ollama, Transformers, OpenAI, ChatGPT
- **Cloud & DevOps:** AWS (Lambda, S3, Redshift, SageMaker, EMR, Transfer Family, EventBridge, SNS, SQS, EC2, VPC, Cloudwatch, Secret Manager, Quicksight), Docker, Git, GitHub Actions
- **Automation and Scrapping:** Selenium, BeautifulSoup, Scrapy
- **Database:** SQL Server, PostgreSQL, MongoDB, Pinecone, Chromadb, FAISS
- **Tools and IDEs:** VS Code, Google Colab, Jupyter Notebook, PyCharm, Visual Studio, Wireshark, Postman, Dbeaver, RStudio, SSRS, SSIS, PowerBI, Anaconda Navigator